



BOILEAU GUILLAUME

Space Data Operations & Processing Engineer | Python, Linux, pipelines, monitoring & anomaly analysis

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EXPERIENCE

Software Engineer - System Integration, Data Flows & Operational Monitoring

VEV Platform Services France

2025 - 2026

Valbonne, France

Context: Development and integration of software services for an EV charging CPMS platform connected to operational infrastructure, charging stations, data flows and hardware systems.

Objective: Support reliable operational data-flow continuity, monitoring, incident analysis and software/system integration in an industrial environment.

- **Operational systems:** Contributed to the development, deployment support and maintenance of software services connected to operational infrastructure and field equipment.
- **Data flows:** Implemented, monitored and corrected FTP-based operational data exchange services involving structured data and recurring processing actions.
- **Monitoring & reliability:** Used operational logs, OpenSearch and Sentry to investigate incidents, follow system behaviour and support service reliability.
- **Anomaly analysis:** Analysed issues involving software behaviour, operational data exchanges, interface continuity and hardware-connected systems.
- **Structured data:** Manipulated SQL, relational concepts and JSON-based objects/configurations for daily technical actions and workflows.
- **System integration:** Worked on interactions between platform services, connected charging stations, data exchanges, electrical circuits, communication buses and operational behaviours.
- **Technical reporting:** Documented incidents, technical findings, corrective actions and operational observations for internal teams.
- **Industrial environment:** Worked with software, operations and technical stakeholders to clarify issues, follow priorities and support iterative delivery.
- **Tooling:** Used Git-based workflows, Zenhub, Confluence, OpenSearch and Sentry for delivery follow-up, diagnostics and technical documentation.

Senior R&D Engineer - Space Data Processing, Validation & Technical Coordination

CNES, Laboratoire Lagrange, Observatoire de la Côte d'Azur

2023 - 2025

Nice, France

Context: Engineering and data-analysis activities for the LISA ESA/NASA space mission, involving complex software chains, simulation outputs, data products, validation workflows and international coordination.

Objective: Develop, integrate and validate Python-based data processing workflows under strong consistency, traceability, performance and verification constraints.

- **Space data processing:** Developed Python workflows for the processing, comparison and validation of complex simulation outputs and space-mission data products.
- **Pipeline development:** Developed CATpyLISA, a Python platform for large-scale model integration, comparison, validation and technical review. [GitLab: CATpyLISA](#)
- **Operational logic:** Structured repeatable processing sequences, reruns, consistency checks, data-product consolidation and validation-oriented diagnostics.
- **Reprocessing workflows:** Supported iterative processing and re-analysis of heterogeneous outputs to assess robustness, consistency and impact of configuration changes.
- **Data validation:** Ensured traceability of inputs, configurations, outputs, assumptions, metadata and validation results.
- **Anomaly investigation:** Analysed discrepancies between models, outputs and expected behaviours to identify causes, limitations and corrective directions.
- **Performance analysis:** Analysed sensitivity to modelling assumptions, numerical parameters, data-processing choices and pipeline configurations.
- **Signal processing:** Analysed weak, noisy and superposed signals using spectral methods, signal/noise separation logic and performance metrics.
- **Technical reporting:** Produced analysis notes, validation summaries, technical documentation and review material for contributors and project stakeholders.
- **Coordination:** Coordinated contributors, supervised interns and supported technical activities across multiple stakeholders in an international environment.

Data Scientist - Noise Diagnostics, Data Analysis & Performance Validation

Universiteit Antwerpen, Belgium

2021 - 2023

Antwerp, Belgium

Context: Data analysis and performance studies for precision instruments in a high-requirement international environment linked to gravitational-wave detector performance.

Objective: Identify, characterize and mitigate factors affecting measurement quality and system sensitivity using statistical modelling, signal processing and validation-oriented methods.

- **Data processing:** Analysed measurement and simulation outputs to identify abnormal behaviours, noise contributions and degradation sources.

PROFILE

Engineer at the interface between space data processing, operational workflows, software integration, validation and anomaly analysis. Background developed through ESA/NASA space mission activities, scientific data pipelines, precision instrumentation and industrial operational systems.

Experienced in **Python, Linux, Shell/Bash, SQL, Docker**, structured data handling, JSON workflows, data processing pipelines, validation logic, monitoring, technical reporting and system reliability analysis.

For a **Space Data Operations & Processing Engineer** role, I bring a strong engineering profile: developing and operating data workflows, analysing anomalies, validating data products, supporting reprocessing activities, documenting results and working across software, data, operations and system constraints.

FIT FOR SPACE DATA OPERATIONS

- Strong experience in Python-based data processing pipelines for space-mission and scientific data products.
- Solid Linux, Bash, Git, Docker and reproducible workflow background.
- Experience with SQL, relational database concepts and daily manipulation of JSON-based structured data.
- Strong ability to analyse anomalies, investigate discrepancies and contribute to corrective actions.
- Experience with operational monitoring, logs, OpenSearch, Sentry and production-like incident analysis.
- Background in validation, traceability, technical documentation and performance-oriented reporting.
- Strong fit for international technical environments requiring autonomy, rigor, analytical thinking and technical English.
- Space-domain experience from LISA ESA/NASA; directly transferable to satellite data processing and operational data workflows.

TEAM MANAGEMENT

- Supervision of **3 Master's thesis interns (M2)** during software, modelling and data-analysis activities.
- Coordination of technical activities involving **research engineers and technicians** on a **data processing and validation pipeline** for the **LISA space mission**.
- Experience organizing tasks, supporting contributors, reviewing outputs and maintaining alignment across collaborative technical developments.

LANGUAGES

English ● ● ● ● ●

Spanish ● ● ● ● ●

EDUCATION

PhD in Physics

Université Côte d'Azur

2018 - 2021

Nice, France

Selective Graduate Program in Fundamental Physics (Magistère)

- **Signal processing:** Analysed instrumental and environmental effects affecting measurement quality, sensitivity and stability.
- **Noise diagnostics:** Developed methods for spectral analysis, noise characterization, sensitivity studies and cross-sensor correlations.
- **Performance monitoring:** Identified contributors to performance degradation and supported prioritisation of correction or mitigation actions.
- **Reproducibility:** Produced traceable, reproducible and documented analyses for international scientific and engineering stakeholders.
- **Technical communication:** Communicated complex results through structured reporting, presentations and collaborative review.

Junior R&D Engineer - Simulation, Signal Analysis & Scientific Computing

ARTEMIS, Observatoire de la Côte d'Azur

📅 2018 – 2021

📍 Nice, France

Context: Data-analysis and simulation activities for gravitational-wave mission preparation, involving weak-signal detection, modelling, simulation and noise characterization.

Objective: Develop robust analysis methods for complex signals and support technical investigations in demanding system environments.

- **Simulation workflows:** Built simulation approaches for complex signal environments, uncertainty studies and large-scale datasets.
- **Signal analysis:** Developed methods for the separation of overlapping low-SNR signals in complex datasets.
- **Algorithm assessment:** Compared simulated outputs, model behaviours and analysis results to assess consistency, robustness and performance limitations.
- **Technical investigations:** Investigated noise sources through cross-sensor correlation analyses in diagnostic contexts.
- **Scientific computing:** Developed strong expertise in Python-based analysis, modelling, documentation and reproducible workflows.

PROFESSIONAL TRAINING

Supervised Machine Learning (Regression & Classification)

DeepLearning.AI - Andrew Ng

📅 2020

📍 Coursera

Recherche reproducible : principes méthodologiques pour une science transparente

FUN MOOC – Inria

📅 2025

📍 France Université Numérique

Continuous Professional Training – Software, Data, AI and Systems

OpenClassrooms

📅 2024 – 2026

Python, SQL, Machine Learning, AI, Linux, JavaScript, TypeScript, Angular, Node.js, Django, REST APIs, C/C++, Java, algorithms, relational databases, software engineering fundamentals and data workflows.

PROJECTS

CNES / LISA – Space Data Processing, Validation & Reprocessing Workflows

ESA/NASA Space Mission - Large-scale International Collaboration

📅 2023 – 2025

📍 Nice, France

Contribution to Python-based data processing, model-comparison and validation workflows for the LISA space mission within a large international engineering and scientific collaboration.

Space data relevance: Processing of heterogeneous simulation outputs, consolidation of data products, traceability of configurations, validation of results and technical review.

Operations relevance: Repeatable processing sequences, reruns, consistency checks, anomaly investigation, structured outputs and reporting for stakeholders.

Critical systems relevance: Mission-level performance analysis, validation evidence, rigorous documentation and work in a high-constraint international aerospace context.

Einstein Telescope / LIGO–Virgo–KAGRA – Data Diagnostics & Detector Performance

International Collaborations

📅 2021 – 2023

📍 Antwerp, Belgium

Contribution to collaborative developments in data analysis, signal characterization, statistical diagnostics and performance investigations within large-scale international scientific and engineering collaborations.

Role: Development of analysis tools, contribution to technical studies, reporting and collaborative review in high-standard environments.

System impact: Studies on environmental and instrumental noise sources supporting the identification of performance limitations and design constraints for next-generation gravitational-wave observatories.

VEV – Operational Data Flows, Monitoring & System Integration

EV Charging Infrastructure / CPMS Platform

📅 2025 – 2026

📍 Valbonne, France

Université Paris-Saclay

📅 2016 – 2018

📍 Orsay, France

MSc in Astronomy and Astrophysics

Observatoire de Paris / Université Paris-Saclay

📅 2017 – 2018

📍 Paris, France

BSc in Fundamental Physics

Université Paris-Saclay

📅 2015 – 2016

📍 Orsay, France

Contribution to software development, operational diagnostics and hardware/software interface analysis for a CPMS platform connected to EV charging infrastructure.

Operational relevance: Monitoring of data exchanges, incident analysis, logs investigation, OpenSearch/Sentry usage, software reliability, JSON/SQL handling and operational traceability.

STRENGTHS

Space data processing & operations

- Space Data Processing
- Satellite Data
- Data Operations
- Data Processing Pipelines
- Reprocessing Workflows
- Operational Workflows
- Pipeline Automation
- Data Validation
- Data Product Consolidation
- Processing Chains
- Workflow Reliability
- Operational Reporting
- Technical Documentation
- Technical English

Programming, Linux & data tools

- Python
- Shell/Bash
- Linux
- SQL
- Relational Databases
- PostgreSQL Basics
- JSON
- Structured Data
- Pandas
- NumPy
- Scikit-Learn
- Jupyter
- Git
- GitLab
- GitHub
- Docker
- CI/CD
- REST API
- TypeScript
- Angular
- Node.js
- Java
- C
- C++

Monitoring, reliability & anomaly analysis

- Operational Monitoring
- OpenSearch
- Sentry
- Log Analysis
- Incident Analysis
- Anomaly Analysis
- Corrective Actions
- Root-Cause Analysis
- System Reliability
- Performance Monitoring
- Data-Flow Monitoring
- Issue Investigation
- Robustness
- Problem Solving
- Analytical Thinking

Signal, diagnostics & complex data

- Signal Processing
- Weak Signal Detection
- Low-SNR Analysis
- Noise Modelling
- Noise Diagnostics
- Spectral Analysis
- Cross-Sensor Correlation
- Simulation Workflows
- Synthetic Data
- Uncertainty Analysis
- Sensitivity Analysis
- Performance Assessment
- Complex Data Analysis
- Scientific Computing

Validation, quality & critical systems

- System Validation
- IVVQ
- V-Cycle
- Verification
- Validation Evidence
- Traceability
- Requirements Awareness
- Technical Evidence
- Data Reliability
- Quality Checks
- Consistency Checks
- Risk Identification
- Critical Systems
- Aerospace Context
- Space Systems

Distributed computing, delivery & collaboration

- HPC
- Slurm
- HTCondor
- Distributed Computing
- Containerized Environments
- Cloud Awareness
- Technical Workshops
- Technical Reporting
- Knowledge Sharing
- Stakeholder Alignment
- Cross-Functional Coordination
- Agile Environment
- Confluence
- JIRA
- Zenhub
- Autonomy
- Rigor
- Team Player